

Build and Configure a Secure Linux Server for a Startup Company

Master Class Program on Cloud Computing and DevOps
Assignment – 1, Week 1

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1 Introduction

As part of the Cloud Computing and DevOps training program, this assignment simulates the role of a Junior Linux & DevOps Engineer at ABC Tech Solutions Pvt. Ltd. The objective was to configure a secure Linux server for organizational use, covering user and group management, directory permissions, SSH remote access, FTP file transfer, Apache web hosting, and Bash automation. All tasks were performed on an Ubuntu Server VM set up in VirtualBox.

2 Task 1: User and Group Management

2.1 Objective

Create user accounts `developer1`, `developer2`, `tester1`, and `manager`, create groups `developers` and `testers`, and assign users to their respective groups.

2.2 Commands Executed

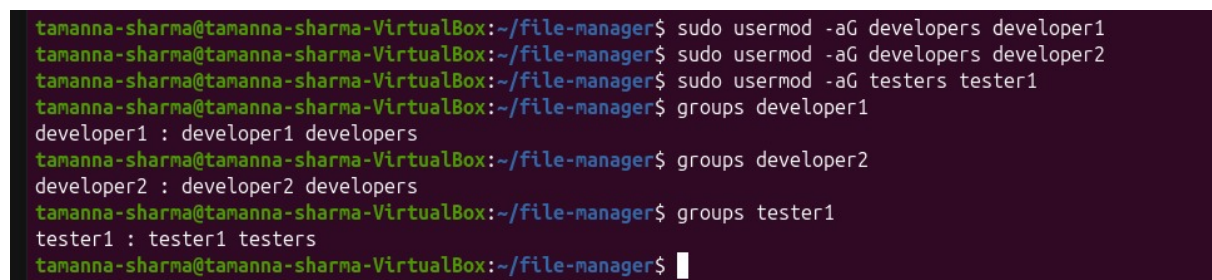
```
sudo useradd -m developer1
sudo useradd -m developer2
sudo useradd -m tester1
sudo useradd -m manager

sudo groupadd developers
sudo groupadd testers

sudo usermod -aG developers developer1
sudo usermod -aG developers developer2
sudo usermod -aG testers tester1

groups developer1
groups developer2
groups tester1
```

2.3 Screenshot



```
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ sudo usermod -aG developers developer1
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ sudo usermod -aG developers developer2
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ sudo usermod -aG testers tester1
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ groups developer1
developer1 : developer1 developers
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ groups developer2
developer2 : developer2 developers
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ groups tester1
tester1 : tester1 testers
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$
```

Figure 1: Assigning users to groups and verifying group membership

2.4 Observation

The `groups` command confirms that `developer1` and `developer2` belong to the `developers` group, while `tester1` belongs to the `testers` group, as required.

3 Task 2: Directory Structure and Permissions

3.1 Objective

Create the directory hierarchy `/company`, `/company/developers`, `/company/testers`, and `/company/backups`, and configure permissions so that developers can access only their directory, testers can access only their directory, and the backup directory is restricted to the administrator.

3.2 Commands Executed

```
sudo mkdir -p /company/developers /company/testers /company/backups

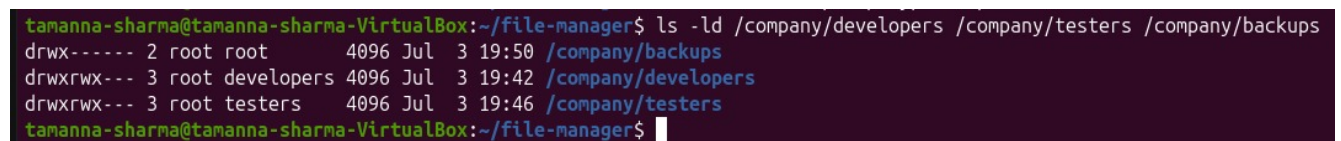
sudo chown root:developers /company/developers
sudo chmod 770 /company/developers

sudo chown root:testers /company/testers
sudo chmod 770 /company/testers

sudo chown root:root /company/backups
sudo chmod 700 /company/backups

ls -ld /company/developers /company/testers /company/backups
```

3.3 Screenshot

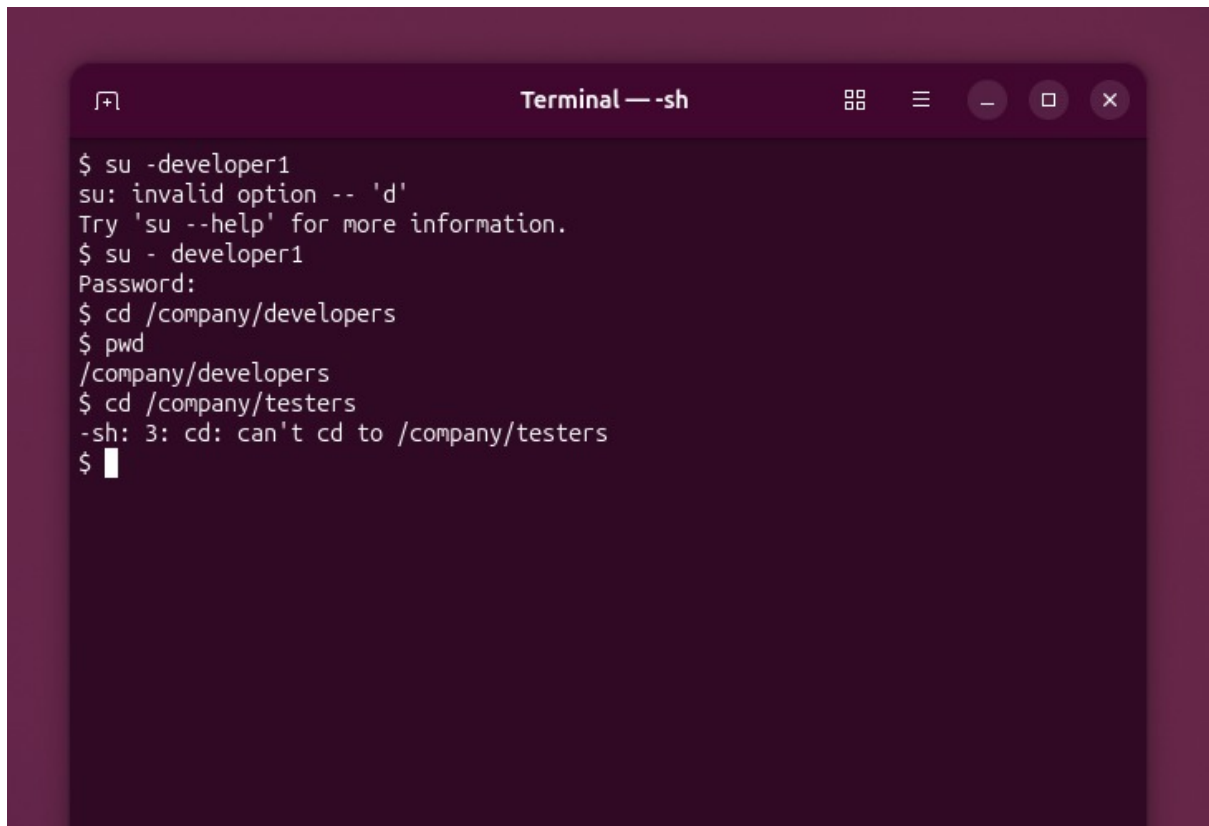


```
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ ls -ld /company/developers /company/testers /company/backups
drwx----- 2 root root      4096 Jul  3 19:50 /company/backups
drwxrwx--- 3 root developers 4096 Jul  3 19:42 /company/developers
drwxrwx--- 3 root testers   4096 Jul  3 19:46 /company/testers
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$
```

Figure 2: Directory permissions verified using `ls -ld`

3.4 Verification

Switching to `developer1` confirms access to `/company/developers` but denies access to `/company/testers`, proving that the permission boundaries are correctly enforced.



```
Terminal — -sh
$ su -developer1
su: invalid option -- 'd'
Try 'su --help' for more information.
$ su - developer1
Password:
$ cd /company/developers
$ pwd
/company/developers
$ cd /company/testers
-sh: 3: cd: can't cd to /company/testers
$
```

Figure 3: developer1 can access developers directory but not testers directory

3.5 Observation

/company/backups has permission `drwx-----` owned by root, so only the administrator can access it. The `developers` and `testers` directories use group ownership with `770` permissions, restricting access to members of the respective group only.

4 Task 3: SSH Configuration

4.1 Objective

Configure SSH on the server and demonstrate a successful remote login from another Linux system.

4.2 Commands Executed

```
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh
sudo systemctl status ssh

hostname -I

ssh developer1@10.0.2.15
whoami
```

```
hostname
exit
```

4.3 Screenshots

```
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-03 20:05:26 IST; 2min 12s ago
     Invocation: 1e2c6bf5df694bf4933e38b48e76c122
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1309 (sshd)
       Tasks: 1 (limit: 1917)
      Memory: 1.1M (peak: 2.4M, swap: 100K, swap peak: 100K)
         CPU: 65ms
    CGroup: /system.slice/ssh.service
            └─1309 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 03 20:05:25 tamanna-sharma-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jul 03 20:05:26 tamanna-sharma-VirtualBox sshd[1309]: Server listening on 0.0.0.0 port 22.
Jul 03 20:05:26 tamanna-sharma-VirtualBox sshd[1309]: Server listening on :: port 22.
Jul 03 20:05:26 tamanna-sharma-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$
```

Figure 4: SSH service active and running

```
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ hostname -I
10.0.2.15 fd17:625c:f037:2:a00:27ff:fe1e:e70a
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$
```

Figure 5: Server IP address obtained using hostname -I

```
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ ssh developer1@10.0.2.15
developer1@10.0.2.15's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-27-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

9 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Figure 6: Successful SSH login as developer1

```
$ whoami
developer1
$ hostname
tamanna-sharma-VirtualBox
$ exit
Connection to 10.0.2.15 closed.
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$
```

Figure 7: Verifying logged-in user and hostname over SSH session, then exiting

4.4 Recorded Details

- IP Address of server: 10.0.2.15
- Username used for login: developer1

5 Task 4: FTP Configuration

5.1 Objective

Configure an FTP server and transfer files between two systems, including upload and download, followed by verification.

5.2 Commands Executed

```
sudo apt install vsftpd -y
sudo systemctl enable vsftpd
sudo systemctl start vsftpd
sudo systemctl status vsftpd

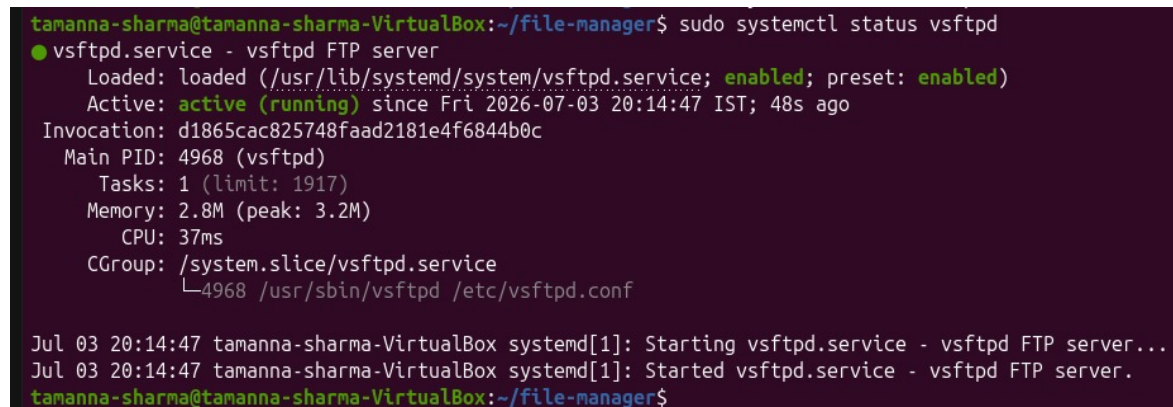
sudo nano /etc/vsftpd.conf
# local_enable=YES
# write_enable=YES
sudo systemctl restart vsftpd

ftp 10.0.2.15
put testfile.txt

ftp localhost
lcd /home/tamanna-sharma
get textfile.txt
get textfile.txt textfile_downloaded.txt

ls -l textfile.txt textfile_downloaded.txt
diff textfile.txt textfile_downloaded.txt
```

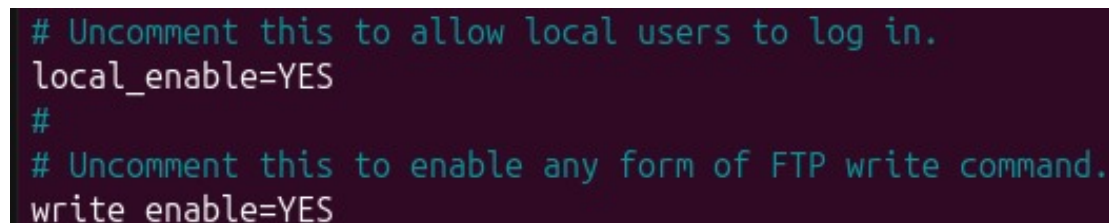
5.3 Screenshots



```
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-03 20:14:47 IST; 48s ago
     Invocation: d1865cac825748faad2181e4f6844b0c
       Main PID: 4968 (vsftpd)
         Tasks: 1 (limit: 1917)
        Memory: 2.8M (peak: 3.2M)
           CPU: 37ms
        CGroup: /system.slice/vsftpd.service
                └─4968 /usr/sbin/vsftpd /etc/vsftpd.conf

Jul 03 20:14:47 tamanna-sharma-VirtualBox systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Jul 03 20:14:47 tamanna-sharma-VirtualBox systemd[1]: Started vsftpd.service - vsftpd FTP server.
tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$
```

Figure 8: vsftpd service active and running



```
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
```

Figure 9: vsftpd.conf configured with local_enable and write_enable set to YES


```

tamanna-sharma@tamanna-sharma-VirtualBox:~/file-manager$ ftp 10.0.2.15
Connected to 10.0.2.15.
220 (vsFTPD 3.0.5)
Name (10.0.2.15:tamanna-sharma): developer1
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>

```

Figure 10: Successful FTP login as developer1

```

ftp> put testfile.txt
local: testfile.txt remote: testfile.txt
229 Entering Extended Passive Mode (|||8546|)
150 Ok to send data.
100% |*****| 56 710.22 KiB/s 00:00 ETA
226 Transfer complete.
56 bytes sent in 00:00 (43.54 KiB/s)
ftp>

```

Figure 11: File uploaded to the server using the put command

```

tamanna-sharma@tamanna-sharma-VirtualBox:~$ ftp localhost
Connected to localhost.
220 (vsFTPD 3.0.5)
Name (localhost:tamanna-sharma): developer1
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> lcd /home/tamanna-sharma
Local directory now: /home/tamanna-sharma
ftp> lpwd
Local directory: /home/tamanna-sharma
ftp> get testfile.txt
local: testfile.txt remote: testfile.txt
229 Entering Extended Passive Mode (|||57111|)
150 Opening BINARY mode data connection for testfile.txt (56 bytes).
100% |*****| 56 331.43 KiB/s 00:00 ETA
226 Transfer complete.
56 bytes received in 00:00 (27.53 KiB/s)
ftp> get testfile.txt testfile_downloaded.txt
local: testfile_downloaded.txt remote: testfile.txt
229 Entering Extended Passive Mode (|||32563|)
150 Opening BINARY mode data connection for testfile.txt (56 bytes).
100% |*****| 56 193.92 KiB/s 00:00 ETA
226 Transfer complete.
56 bytes received in 00:00 (105.98 KiB/s)
ftp> bye
221 Goodbye.
tamanna-sharma@tamanna-sharma-VirtualBox:~$

```

Figure 12: Files downloaded from the server using the get command

```

221 Goodbye.
tamanna-sharma@tamanna-sharma-VirtualBox:~$ ls -l testfile.txt testfile_downloaded.txt
-rw-rw-r-- 1 tamanna-sharma tamanna-sharma 56 Jul  4 2026 testfile.txt
-rw-rw-r-- 1 tamanna-sharma tamanna-sharma 56 Jul  4 2026 testfile_downloaded.txt
tamanna-sharma@tamanna-sharma-VirtualBox:~$ diff testfile.txt testfile_downloaded.txt
tamanna-sharma@tamanna-sharma-VirtualBox:~$

```

Figure 13: Verifying the downloaded file is identical to the original using diff

5.4 Observation

The `diff` command produced no output, confirming that `textfile.txt` and `textfile_downloaded.txt` are byte-for-byte identical, verifying a successful and lossless file transfer.

6 Task 5: Web Server Deployment

6.1 Objective

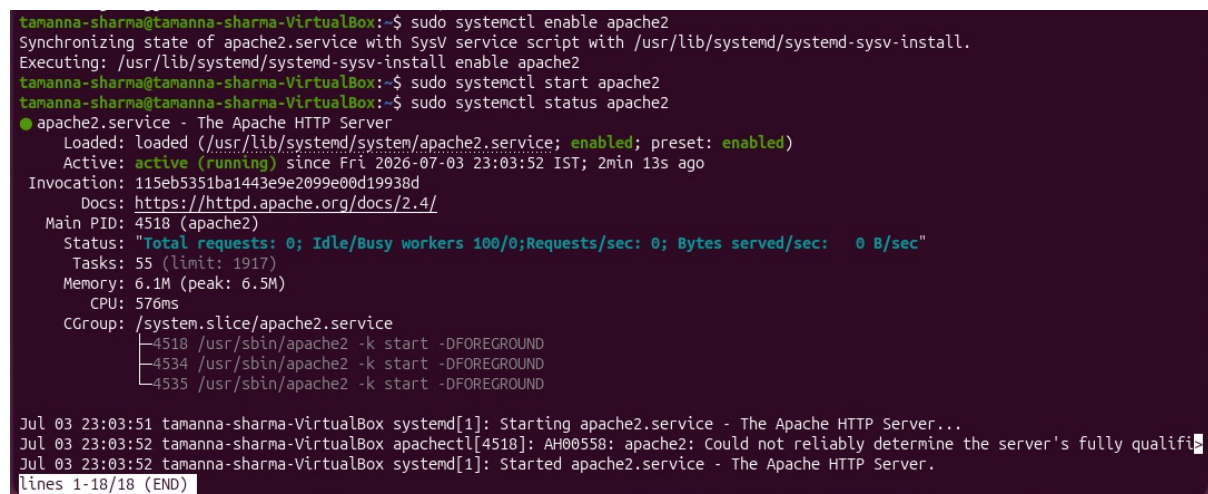
Install and configure the Apache web server and host a webpage containing the project title, name, roll number, branch, current date and time, and a welcome message.

6.2 Commands Executed

```
sudo apt install apache2 -y
sudo systemctl enable apache2
sudo systemctl start apache2
sudo systemctl status apache2

sudo nano /var/www/html/index.html
```

6.3 Screenshots

A terminal window screenshot showing the installation and status of the Apache2 service. The user is in a VirtualBox environment. The commands executed are: `sudo systemctl enable apache2`, `sudo systemctl start apache2`, and `sudo systemctl status apache2`. The output shows that the service is loaded, enabled, and active (running). It also displays system statistics like requests per second, memory usage, and CPU usage. At the bottom, there are system logs indicating the service was started successfully.

```
tamanna-sharma@tamanna-sharma-VirtualBox:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
tamanna-sharma@tamanna-sharma-VirtualBox:~$ sudo systemctl start apache2
tamanna-sharma@tamanna-sharma-VirtualBox:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-03 23:03:52 IST; 2min 13s ago
 Invocation: 115eb5351ba1443e9e2099e00d19938d
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 4518 (apache2)
  Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 55 (limit: 1917)
   Memory: 6.1M (peak: 6.5M)
      CPU: 576ms
   CGroup: /system.slice/apache2.service
           └─4518 /usr/sbin/apache2 -k start -DFOREGROUND
             └─4534 /usr/sbin/apache2 -k start -DFOREGROUND
               └─4535 /usr/sbin/apache2 -k start -DFOREGROUND

Jul 03 23:03:51 tamanna-sharma-VirtualBox systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jul 03 23:03:52 tamanna-sharma-VirtualBox apachectl[4518]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the mod_ssl module docs
Jul 03 23:03:52 tamanna-sharma-VirtualBox systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-18/18 (END)
```

Figure 14: Apache2 service enabled, started, and running

Build and Configure a Secure Linux Server For a Startup Company

Name: Tamanna Sharma

Roll Number: 2302697

Branch: Computer Science And Engineering

Current Date & Time: 7/4/2026, 2:06:30 PM

Welcome to XYZ Tech Solutions Pvt. Ltd. server!

Figure 15: Webpage accessed via the server's IP address in a browser

6.4 Observation

The webpage was accessed successfully using the server's IP address, and it correctly displays the project title, name, roll number, branch, current date and time, and a welcome message for visitors.

7 Task 6: Bash Automation Script

7.1 Objective

Develop a Bash script named `server_health.sh` that displays the current date and time, logged-in user, hostname, IP address, disk usage, memory usage, CPU load, and system uptime in a well-formatted manner.

7.2 Script: `server_health.sh`

```
#!/bin/bash

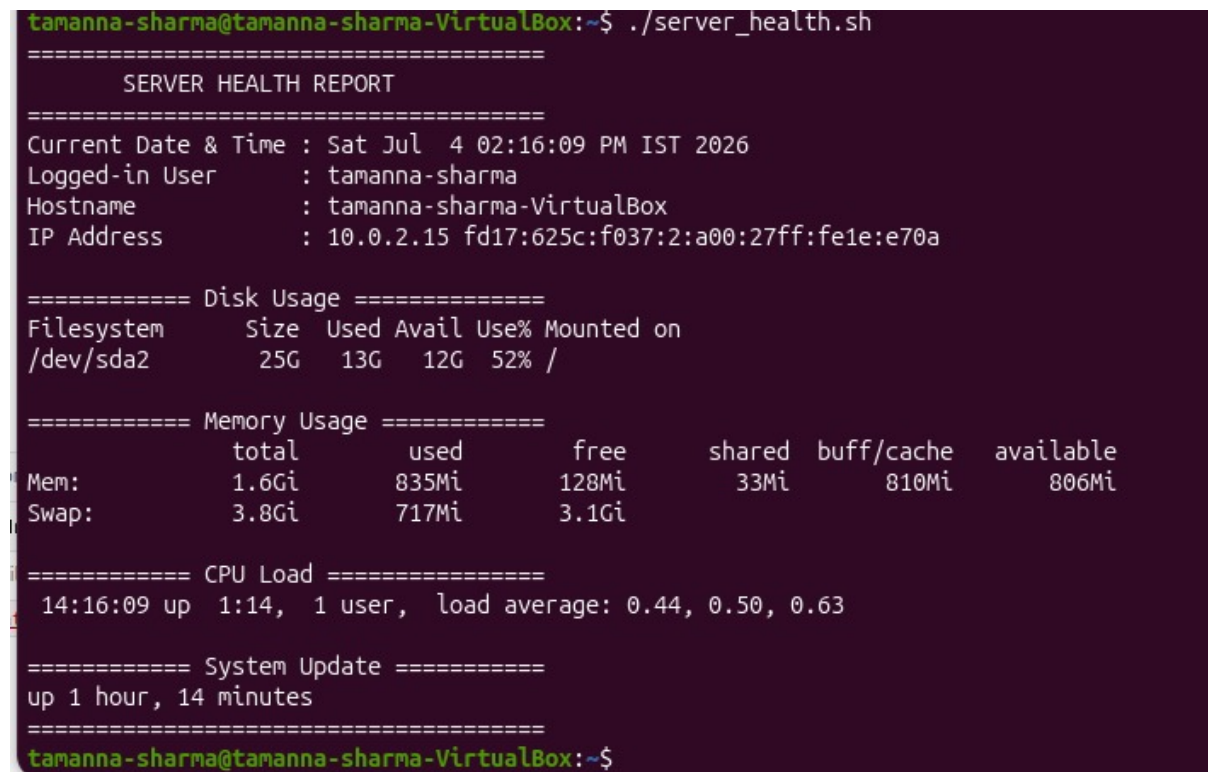
echo "=====
echo " SERVER HEALTH REPORT"
echo "=====
echo "Current Date & Time : $(date '+%a %b %d %I:%M:%S %p %Z %Y')
echo "Logged-in User : $(whoami)"
echo "Hostname : $(hostname)"
echo "IP Address : $(hostname -I)"
echo
echo "===== Disk Usage =====
df -h /
echo
echo "===== Memory Usage =====
free -h
echo
echo "===== CPU Load =====
uptime
echo
echo "===== System Uptime ====="
```

```
uptime -p
echo "=====
```

7.3 Commands Executed

```
chmod +x server_health.sh
./server_health.sh
```

7.4 Screenshot

A terminal window showing the output of the server_health.sh script. The output is a structured report with sections for system information, disk usage, memory usage, CPU load, and system uptime. The terminal background is dark purple with light green text.

```
tamanna-sharma@tamanna-sharma-VirtualBox:~$ ./server_health.sh
=====
SERVER HEALTH REPORT
=====
Current Date & Time : Sat Jul  4 02:16:09 PM IST 2026
Logged-in User      : tamanna-sharma
Hostname            : tamanna-sharma-VirtualBox
IP Address          : 10.0.2.15 fd17:625c:f037:2:a00:27ff:fe1e:e70a

===== Disk Usage =====
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2       25G   13G   12G   52% /

===== Memory Usage =====
              total        used         free      shared  buff/cache   available
Mem:          1.6Gi         835Mi        128Mi         33Mi       810Mi       806Mi
Swap:         3.8Gi         717Mi         3.1Gi

===== CPU Load =====
14:16:09 up 1:14, 1 user, load average: 0.44, 0.50, 0.63

===== System Update =====
up 1 hour, 14 minutes
=====
tamanna-sharma@tamanna-sharma-VirtualBox:~$
```

Figure 16: Output of server_health.sh showing system health report

7.5 Observation

The script successfully consolidates key system health metrics – date and time, user, hostname, IP address, disk usage, memory usage, CPU load, and uptime – into a single, readable report.

8 Challenges Encountered

- Initially attempted `su -developer1` without a space, which resulted in an invalid option error; this was resolved by using `su - developer1`.
- Faced a permission issue while switching users and navigating to restricted directories, which helped confirm that the directory permissions were working as intended.

- Had to correct the `vsftpd.conf` file to uncomment `local_enable=YES` and `write_enable=YES` before FTP uploads would work.
- Apache2 displayed a warning about not being able to reliably determine the server's fully qualified domain name, which did not affect functionality but was noted during service startup.

9 Learning Outcomes

- Gained hands-on experience configuring Linux users, groups, and directory-level permissions to enforce access control.
- Learned to set up and verify SSH for secure remote administration between two systems.
- Configured an FTP server and understood the difference between passive mode data transfer and verifying file integrity using `diff`.
- Deployed a working website using the Apache web server and understood basic web server configuration.
- Practiced Bash scripting to automate the collection and formatted display of system health metrics.
- Developed a better understanding of how these individual components (users, permissions, SSH, FTP, web hosting, automation) work together in a real server administration workflow.

10 Conclusion

This assignment provided practical, end-to-end exposure to Linux server administration tasks that are directly relevant to a DevOps role. By creating users and groups, enforcing directory permissions, configuring SSH and FTP, deploying a web server, and automating health checks with a Bash script, the server was successfully configured to meet the requirements of a startup company scenario. All tasks were completed and verified with appropriate screenshots, and the exercise strengthened both command-line proficiency and understanding of secure server configuration practices.